

# A normalized discrete kernel disproves the universal probabilistic representer conjecture

Candidate counterexample to Conjecture 4.1 of arXiv:2409.11679

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**Status: candidate counterexample, likely valid; full disproof as stated.** There is a normalized kernel on a countable discrete Polish space and a smooth, strictly convex cost satisfying every hypothesis of Theorem 2.6 for which the unique optimizer is the vector  $(1/n) \in \ell^2 \setminus \ell^1$ . Kernel embeddings of finite signed measures have exactly  $\ell^1$  coefficients, so the optimizer has no measure representation.

## Source conjecture

Chen and Wang’s Conjecture 4.1 asks whether the unique optimizer  $\hat{f}$  in their regularized approximation theorem must have the form

$$\hat{f} = \int_X \phi(x) d\nu(x)$$

for a signed or complex measure  $\nu$  supported on  $\text{supp } \mu$ , provided every such measure  $\xi$  satisfies

$$\int_X \|\phi(x)\|_{\mathcal{H}(K)} d|\xi|(x) < \infty.$$

The original statement appears on PDF page 13 of the source paper [1, Conjecture 4.1] and is reproduced below.

conclude that  $f \in \Omega$ . If  $\mu$  is finitely supported or  $\mathcal{H}(K)$  is finite-dimensional, the set  $\mathcal{A}$  is automatically closed and thus  $\hat{f} \in \Omega = \mathcal{A}$ .  $\square$

#### 4. DISCUSSIONS ON THE REPRESENTER THEOREM

The preferable result in the probabilistic representer theorem (Theorem 3.2) is that the minimizer  $\hat{f}$  is the kernel embedding of some  $\mathbb{F}$ -measure  $\nu$  such that  $\hat{f}$  can be represented directly by the  $\mathbb{F}$ -measure  $\nu$ , instead of an approximating sequence  $\{\nu_n\}$ . We conjecture that such a desirable closed form holds merely under the assumption 3.1:

**Conjecture 4.1.** *Let  $\hat{f}$  be the unique optimizer in Theorem 2.6. Assume the probability measure  $\mu$  satisfies that for any  $\mathbb{F}$ -measure  $\xi$  on  $X$  with  $\text{supp}(\xi) \subset \text{supp}(\mu)$ , the following holds:*

$$\int_X \|\phi(x)\|_{\mathcal{H}(K)} d|\xi|(x) < +\infty.$$

*Then there exists an  $\mathbb{F}$ -measure  $\nu$  on  $X$  with  $\text{supp}(\nu) \subset \text{supp}(\mu)$  such that*

$$\hat{f} = \int_X \phi(x) d\nu(x).$$

Once this conjecture is true, the optimizer of Problem 1.3 has a measure representation form, which makes the probabilistic approximation problem become a sampling problem from the perspective of numerical applications: sample the probability measure  $\mu$  to get an  $\mathbb{F}$ -measure  $\nu = \sum_{i=1}^m w_i \delta_{y_i}$  on  $X$  and a candidate  $\int_X \phi(x) d\nu(x)$  for the optimizer, and then update  $\nu$  to make the regularized loss function in Problem 1.3 as small as possible. Therefore, this conjecture has potential applications in sparse kernel function approximation and RKHSs play a central

Theorem 2.6 minimizes

$$\int_X c(f(x), g(x)) d\mu(x) + \lambda \|f\|_{\mathcal{H}(K)}^p, \quad (1)$$

where  $\lambda > 0$ ,  $p > 1$ ,  $c(0, g(\cdot)) \in L^1(\mu)$ , and  $c(\cdot, z)$  is lower semicontinuous and convex for each  $z$ . Under these hypotheses the optimizer exists and is unique.

## Proof intuition

For the delta kernel on  $\mathbb{N}$ , the RKHS is  $\ell^2$  and a measure embedding is simply the sequence of its atomic masses. Finite signed measures therefore produce only  $\ell^1$  sequences, although their embeddings are dense in  $\ell^2$ . This is exactly the nonclosed-range phenomenon behind the source's approximating-sequence theorem.

It remains to make a vector in  $\ell^2 \setminus \ell^1$  occur as the unique optimizer without violating the source's integrability assumption. The allowed cost may depend on the target value. We encode at point  $n$  a quadratic loss of weight  $2^n$  and take the sampling mass to be  $2^{-n}$ . These factors cancel in the objective, and completing the square makes  $(1/n)$  the unique optimizer. A smooth global formula for the cost realizes this encoding, so no pathology in the first-variable convexity is involved.

## Counterexample

Let  $X = \mathbb{N} = \{1, 2, \dots\}$  with the discrete topology, let

$$K(m, n) = \delta_{mn}, \quad \mu(\{n\}) = 2^{-n},$$

and work over  $\mathbb{R}$ . Thus  $X$  is Polish,  $\mathcal{H}(K) = \ell^2(\mathbb{N})$ , and the feature map is  $\phi(n) = e_n$ . Set  $p = 2$  and  $\lambda = 1$ . For  $z \in \mathbb{R}$  write

$$q(z) = \sqrt{1 + z^2}, \quad g(n) = \sqrt{n^2 - 1}, \quad c(u, z) = 2^{q(z)} \left( u - \frac{2}{q(z)} \right)^2.$$

**Theorem 1.** *The data above satisfy all assumptions of Theorem 2.6 and of Conjecture 4.1. The unique optimizer of (1) is*

$$\hat{f} = (1, 1/2, 1/3, \dots) \in \ell^2.$$

*There is no finite signed measure  $\nu$  on  $X$  such that  $\hat{f} = \int_X \phi d\nu$ . Consequently Conjecture 4.1 is false.*

*Proof.* The kernel is normalized because  $K(n, n) = 1$ . Hence for every finite signed measure  $\xi$  on  $X$ ,

$$\int_X \|\phi(n)\|_{\ell^2} d|\xi|(n) = \sum_{n=1}^{\infty} |\xi(\{n\})| = |\xi|(X) < \infty.$$

Also every point has positive  $\mu$ -mass, so  $\text{supp } \mu = X$ . This verifies the extra hypothesis of the conjecture.

The functions  $g$  and  $c$  are Borel measurable. In fact  $c$  is continuous on  $\mathbb{R}^2$ , and  $c(\cdot, z)$  is a nonnegative strictly convex quadratic for every  $z$ . Since  $q(g(n)) = n$ ,

$$\int_X c(0, g(n)) d\mu(n) = \sum_{n=1}^{\infty} 2^{-n} 2^n \left( \frac{2}{n} \right)^2 = 4 \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

Thus all hypotheses of Theorem 2.6 hold.

For  $f = (f_n) \in \ell^2$ , the objective is finite and equals

$$\begin{aligned} J(f) &= \sum_{n=1}^{\infty} 2^{-n} c(f_n, g(n)) + \sum_{n=1}^{\infty} f_n^2 \\ &= \sum_{n=1}^{\infty} \left[ \left( f_n - \frac{2}{n} \right)^2 + f_n^2 \right] \\ &= 2 \sum_{n=1}^{\infty} \left( f_n - \frac{1}{n} \right)^2 + 2 \sum_{n=1}^{\infty} \frac{1}{n^2}. \end{aligned}$$

It follows immediately that the unique optimizer is  $\hat{f}_n = 1/n$ . This sequence lies in  $\ell^2$ .

Suppose that a finite signed measure  $\nu$  represented  $\hat{f}$ . On the discrete space  $X$ , put  $a_n = \nu(\{n\})$ . Finiteness of total variation gives

$$\sum_{n=1}^{\infty} |a_n| = |\nu|(X) < \infty. \tag{2}$$

For every  $k$ , the defining identity for the Pettis integral yields

$$\left\langle \int_X e_n d\nu(n), e_k \right\rangle = \int_X \langle e_n, e_k \rangle d\nu(n) = \nu(\{k\}) = a_k.$$

Equality with  $\hat{f}$  would force  $a_k = 1/k$  for all  $k$ , contradicting (2). No representing measure exists.  $\square$

## Sharp mechanism and scope

For this normalized delta kernel, the kernel-mean map from finite signed measures has range exactly  $\ell^1$  inside  $\ell^2$ . Its closure is all of  $\ell^2$ , since finitely supported sequences are dense. Thus the source’s Theorem 3.2 correctly supplies the truncation measures

$$\nu_N = \sum_{n=1}^N \frac{1}{n} \delta_n, \quad \int_X \phi d\nu_N \longrightarrow \hat{f} \quad \text{in } \ell^2,$$

but no single finite signed measure can replace the sequence.

The example disproves the conjecture under its stated general cost hypotheses, even with a normalized kernel and a cost continuous in both variables and strictly convex in the prediction variable. It does not rule out a strengthened representer theorem for narrower loss classes, such as a fixed translation-invariant loss with an integrable pointwise subgradient.

## Verification and novelty bounds

Every hypothesis is checked explicitly above. The completing-square identity and the divergent total variation are exact; the included script checks their finite truncations but is not used in the proof. The source PDF and a full-width crop of Conjecture 4.1 are included in this packet.

On 11 August 2026, bounded searches covered the exact arXiv id and title, “probabilistic representer” with RKHS and measure terms, and the broader nonclosed-range/kernel-mean-embedding formulation, together with all cheap indexes of this run. No later paper answering Conjecture 4.1 and no instance of this counterexample was found. Novelty confidence is moderate, since the  $\ell^1 \subsetneq \ell^2$  obstruction itself is elementary and may be folklore.

Human review should focus on the intended convention that an  $\mathbb{F}$ -measure in the source is a finite signed/complex measure (as its total-variation notation and examples indicate), and on confirming that Theorem 2.6 indeed permits target-dependent weights of the form used here.

## References

- [1] D. Chen and K.-H. Wang, *On the Probabilistic Approximation in Reproducing Kernel Hilbert Spaces*, arXiv:2409.11679, Conjecture 4.1 (PDF p. 13).